

Project Report

Project Title: Reinforcement Learning-Driven Patch Attention for Urban Scene Segmentation

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Introduction

Semantic segmentation is vital for autonomous driving as it classifies every pixel into categories like road, car, and pedestrian. Traditional models treat all pixels equally, which increases computation and reduces accuracy on critical regions. Our project, Reinforcement Learning–Driven Patch Attention, uses an Actor-Critic agent to focus attention on important patches, guiding a U-Net model to achieve better segmentation of roads and vehicles with improved efficiency.

Purpose

I, Setty bhavana, created this project with the goal of:

- Enhance semantic segmentation for urban driving scenes.
 - Focus on critical regions (roads, cars, pedestrians) instead of all pixels equally.
 - Use Reinforcement Learning (Actor-Critic) to assign attention to image patches.
 - Guide a U-Net model with attention-weighted inputs.
 - Achieve higher accuracy and computational efficiency.
 - Support real-world applications like autonomous driving and smart traffic systems.
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Features Implemented

1. Patch-based image division

- Input images are split into spatial patches for focused processing.

2. Actor-Critic RL agent

- Learns to assign attention weights to patches based on importance.

3. Attention-guided U-Net

- Uses weighted patches to prioritize segmentation of critical regions.
4. **Reward function**
 - Designed to maximize mIoU, with higher emphasis on classes like road and vehicles.
 5. **Data augmentation**
 - Applied (flip, resize, color jitter, crop) to improve model generalization.
 6. **Evaluation metrics**
 - Performance measured using pixel accuracy and mean Intersection-over-Union (mIoU).
 7. **Visualization outputs**
 - Attention maps and segmentation masks for clear comparison with ground truth.
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Why I Made This

To improve urban scene segmentation for autonomous driving by focusing on critical regions (roads, vehicles, pedestrians) using reinforcement learning–driven patch attention, achieving higher accuracy and efficiency than traditional models.

Conclusion

This project demonstrates that **reinforcement learning–driven patch attention can significantly improve urban scene** segmentation by focusing on the most critical regions, such as roads, vehicles, and pedestrians. By integrating an Actor-Critic agent with a U-Net model, the approach achieves better accuracy and efficiency compared to traditional segmentation methods. The results highlight the potential of combining reinforcement learning and deep learning for real-world applications like autonomous driving and smart traffic systems.

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